

Assignment 6

Novel Architecture Modification

Strict Protocol Edition · Towards Research-Level Competence

Assignment 6

■ GLOBAL RULES

- Form a hypothesis BEFORE every experiment.
- Explain your reasoning before asking questions.
- After using external help, re-implement from memory.
- Every experiment must include: Hypothesis → Result → Interpretation.

Problem Statement

Design, justify, and rigorously test your own modification to standard attention. You must write a one-page research justification BEFORE implementing anything. This assignment is open-ended -- your hypothesis and ablations define the quality of the work.

- Choose one of: sparse attention patterns, relative position bias, gating mechanisms, or a modification of your own design
- Write a 1-page research justification (hypothesis, expected mechanism, predicted results)
- Implement your modification on top of your best model from previous assignments
- Run a complete ablation: your modification vs. baseline, with all other factors identical
- Evaluate on at least two tasks from previous assignments

Implementation Requirements

1. Implement your chosen modification entirely from scratch -- no copying reference implementations
2. Your baseline must be your own best prior model, not a new reimplementation
3. Run at least 3 ablation variants (e.g., different sparsity levels, gate types, bias ranges)
4. Log all metrics from prior assignments: gradient norms, attention rank, head specialization
5. Write a post-experiment analysis comparing your predictions to actual results

Instrumentation Requirements

1. All instrumentation from Assignments 2-5 must be present
2. Additionally: measure computational cost (FLOPs or wall time) vs. performance tradeoff
3. Log whether your modification changes head specialization behavior
4. Track whether gradient flow improves or degrades with your modification

Failure Analysis (Required)

You MUST document failures. Describe what failed, why you think it failed, test at least two fixes, and evaluate which fix worked and why.

1. Did your modification do what you hypothesized? If not, why not?
2. Is the improvement (if any) consistent across both tasks, or task-specific?
3. What is the simplest version of your modification that still works?
4. Can you break your modification by changing a hyperparameter? Identify the failure mode.

Mathematical Integration

1. Justify your modification mathematically -- what property of standard attention does it address?
2. Analyze the expressivity of your modified attention vs. standard attention
3. Compute spectral norms of any new weight matrices you introduced
4. Relate your modification to at least one published paper (find it yourself, cite it properly)

Research Questions to Answer

- Is your modification a genuine improvement or does it just add parameters?
- What is the inductive bias your modification introduces?
- Under what conditions would you expect your modification to fail?
- If you were submitting this to a workshop, what would the reviewers criticize?

Phase Progression

Phase	Focus
Phase 1	Build the core architecture
Phase 2	Compare with baseline / non-attention models
Phase 3	Study optimization dynamics
Phase 4	Modify loss functions and evaluate

Phase 5	Alter architecture systematically
Phase 6	Design novel training techniques

Document everything like a research paper. Your goal: independently analyze neural architectures, diagnose training failures, and propose novel improvements grounded in theory and empirical evidence.